

DONISHA SMITH

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Research software engineer who builds automated, reproducible pipelines and open-source tools for scientific data

EDUCATION

Florida International University

PhD in Cognitive Neuroscience (M.Sc. en route Aug 2022)

Aug 2025

B.Sc. in Biological Sciences (Cum Laude; QBIC track), Honors College

Apr 2018

EXPERIENCE

Postdoctoral Researcher, Johns Hopkins University School of Medicine

Oct 2025–Present

Advisor: Dr. Kristin Bigos

- Built the lab's PHI-compliant HPC infrastructure, replacing manual workflows with containerized (Apptainer), SLURM-orchestrated pipelines with node-failure recovery, allowing for batch-processing of terabytes of 3D/4D imaging data
- Automated a standardization pipeline for two blinded, multi-session pharmacological studies, reconstructing acquisition metadata from NIfTI headers into a common BIDS schema and deterministically reconciling drug-to-session mappings
- Engineered parallelized, end-to-end pipelines chaining GLM/gPPI modeling with configurable denoising, parametric and nonparametric group-level inference, and cluster identification, with automated HTML QC reports
- Developed CLI and Streamlit web applications enabling non-technical lab staff to independently run multi-session data conversion, event-file generation, and clinical-score PDF extraction
- Designed and deployed a containerized AFNI brain-atlas service (FastAPI, Podman) on Oracle Cloud with a Streamlit frontend for remote brain-region lookup by coordinate, removing the need for local neuroimaging software installations

Graduate Researcher, Florida International University

Aug 2019–Aug 2025

Advisor: Dr. Angela Laird

- Developed and published a k-means clustering pipeline to characterize dynamic brain state transitions in longitudinal fMRI data, identifying distinct transition patterns across timepoints
- Applied linear mixed-effects, structural equation models, and latent profile analyses to multi-session fMRI and behavioral data to identify dissociable brain-behavioral profiles, contributing to two peer-reviewed publications
- Analyzed staff demographic data for the ABCD Study consortium, implementing hypothesis testing and data visualization to inform decisions for a multi-site research task force
- Designed and delivered an R/Tidyverse workshop for 10+ graduate students through a DEI fellowship program, covering data analysis, manipulation, and visualization

Medical Laboratory Scientist I, Florida Department of Health

Jan 2019–Jul 2019

- Conducted confirmatory diagnostic testing for HIV and syphilis for public health surveillance programs

OPEN-SOURCE PROJECTS

BIDS-Aid | [GitHub](#) | [Docs](#)

Oct 2025–Present

- Developed a Python package for post-hoc BIDS conversion of NIfTI datasets lacking source DICOMs, reconstructing JSON sidecars with acquisition metadata derived from image headers
- Built configurable parsers for Presentation and E-Prime behavioral logs, automating timing and accuracy extraction for block and event-related fMRI designs into BIDS-compliant events files
- Achieved a ~5x speedup in NIfTI compression by implementing a Rust extension (PyO3/maturin) for multithreaded compression, distributed as cross-platform abi3 wheels
- Published to PyPI with release-triggered CI/CD and >90% coverage on Python 3.10–3.14 across Linux, macOS, and Windows

NeuroCAPs | [GitHub](#) | [Docs](#) | [Google Colab](#) | [JOSS Paper](#) | [JHU OSPO Catalog](#)

Jan 2024–Present

- Built a Python package for brain state classification using k-means, with multiprocessing for multi-subject timeseries extraction
- Incorporated clustering validation (e.g., Davies-Bouldin, Silhouette), KNN-based spatial interpolation for parcellation coverage, temporal dynamics metrics (e.g., transition probabilities), and integrated visualizations (e.g., surface maps, network alignment)
- Deployed via Docker with headless VTK rendering and CLI/Jupyter interfaces; published to PyPI through release-triggered CI/CD with >90% coverage across multiple OS platforms

- Built a Game Boy Color emulator in Rust, with a CPU passing Blargg's test ROMs and a PPU rendering commercial CGB titles
- Implemented memory bank controllers (MBC1, MBC2, MBC3, MBC5, HuC1) with battery and real-time-clock saves

- Created an R package for ML model evaluation, implementing custom stratified sampling, cross-validation, and nested cross-validation for hyperparameter tuning, with parallel processing support at the fold-level
- Designed a unified interface supporting classification algorithms (e.g., Regularized Logistic Regression, SVM, XGBoost, Neural Networks) while preserving algorithm-specific parameter tuning
- Incorporated automated missing data imputation and multi-metric performance assessment (e.g., precision, recall, F1) with CLI output and visualizations, including ROC and Precision-Recall curves with AUC scores

PUBLICATIONS

[1] **Smith, D. D.**, Bartley, J. E., Peraza, J. A., Bottenhorn, K. L., Nomi, J. S., Uddin, L. Q., Riedel, M. C., Salo, T., Laird, R. W., Pruden, S. M., Sutherland, M. T., Brewe, E., & Laird, A. R. (2025). Dynamic reconfiguration of brain coactivation states associated with active and lecture-based learning of university physics. *Npj Science of Learning*, 10(1), 55. <https://doi.org/10.1038/s41539-025-00348-9>

[2] **Smith, D.** (2025). NeuroCAPs: A Python Package for Performing Co-Activation Patterns Analyses on Resting-State and Task-Based fMRI Data. *Journal of Open Source Software*, 10(112), 8196. <https://doi.org/10.21105/joss.08196>

[3] Pintos Lobo, R., Peraza, J. A., Salo, T., Meca, A., **Smith, D. D.**, Feeney, K. E., Schmarder, K. M., Sutherland, M. T., Gonzalez, R., Musser, E. D., & Laird, A. R. (2025). Social profiles among youth with attention-deficit/hyperactivity disorder (ADHD): Evidence from the ABCD study. *Developmental Cognitive Neuroscience*, 75, 101591. <https://doi.org/10.1016/j.dcn.2025.101591>

[4] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird AR (2023). Task-based attentional and default mode connectivity associated with science and math anxiety profiles among university physics students. *Trends in Neuroscience and Education*. <https://doi.org/10.1016/j.tine.2023.100204>

[5] Lobo, R. P., Bottenhorn, K. L., Riedel, M. C., Toma, A. I., Hare, M. M., **Smith, D. D.**, Moor, A. C., Cowan, I. K., Valdes, J. A., Bartley, J. E., Salo, T., Boevig, E. R., Pankey, B., Sutherland, M. T., Musser, E. D., & Laird, A. R. (2022). Neural systems underlying RDoC social constructs: An activation likelihood estimation meta-analysis. *Neuroscience & Biobehavioral Reviews*. <https://doi.org/10.1016/j.neubiorev.2022.104971>

PRESENTATIONS

[1] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Presented on May 3, 2024, at the Florida Consortium on the Neurobiology of Cognition; Miami, Florida.

[2] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Accepted at the 2024 Graduate Student Appreciation Week at Florida International University; Miami, Florida.

[3] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R. Task-based attention & default mode connectivity linked to STEM anxiety in university students. Presented on July 23, 2023, at the 29th annual meeting of the Organization for Human Brain Mapping; Montréal, Canada.

PEER REVIEW

<i>Journal of Open Source Software</i>	Aug 2025–Present
<i>npj Science of Learning</i>	Jul 2025–Present

FELLOWSHIPS, HONORS & SCHOLARSHIPS

Dissertation Year Fellowship (\$20,000 across two semesters)	Dec 2024–Jun 2025
DEI Doctoral Fellowship (\$3,000 over three semesters)	Aug 2022–Aug 2023
Recipient of Undergraduate NIGMS RISE Fellowship	Jan 2017
FIU Ambassador Scholarship (full-tuition merit scholarship, 2014 cohort)	Aug 2014–May 2018
QBIC Scholarship (\$1,000 per semester)	Aug 2014–May 2018
Florida Academic Scholars Award (highest state merit scholarship tier)	Aug 2014–May 2018

TECHNICAL SKILLS

Programming Languages: Python, R, Rust, Unix Shell Scripting (Bash)

Computing & Development: HPC, CI/CD (Git, GitHub Actions, Docker, Podman, Singularity/Apptainer), Cloud Computing

Neuroimaging Tools: Nilearn, fMRIPrep, AFNI, FSL, E-Prime, Presentation, PsychoPy